

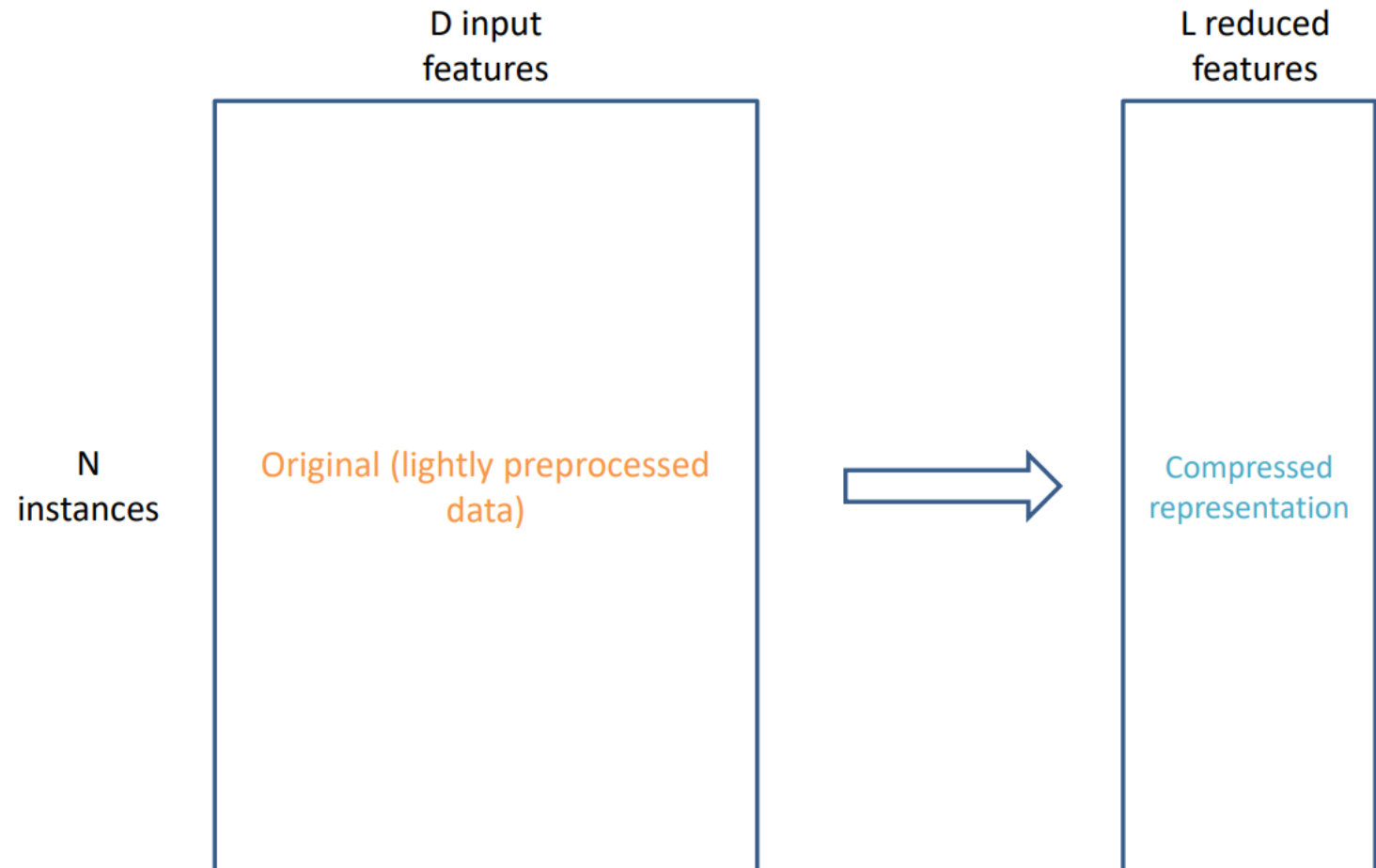
Artificial Intelligence Concepts and Machine Learning Techniques (AI – 103)

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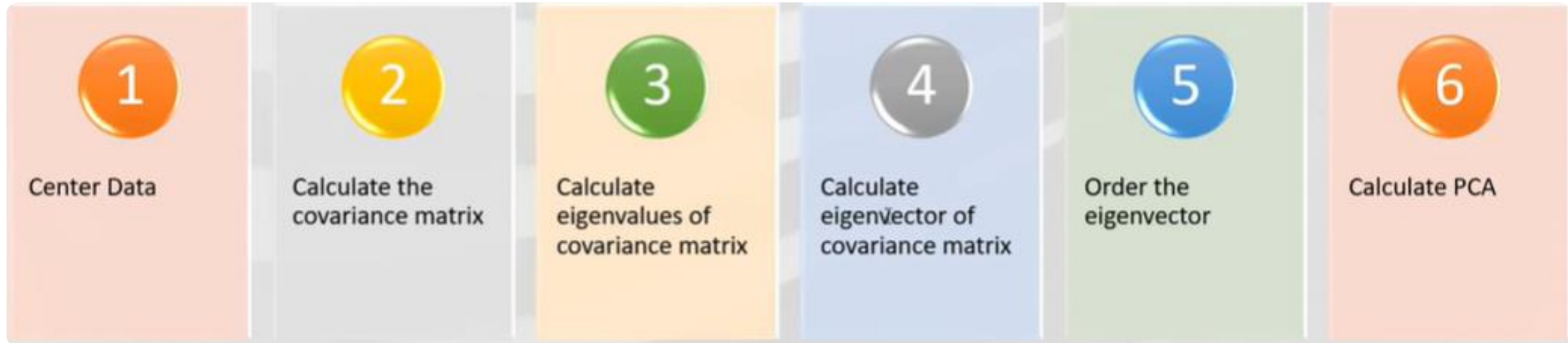
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Dimension Reduction

Dimensionality Reduction



Steps to find PCA



STEP 1: STANDARDIZATION

Purpose:

Different variables often have different ranges.

If we do not standardize them:

- Variables with **larger scales** dominate
- Variables with **smaller ranges** contribute less
- The final PCA results become **biased**

Standardization ensures that **each variable contributes equally** to the analysis.

F1	F2	F3	F4
-1.0695	0.8196	0	-1
0.5347	-1.6393	1.6042	1
-1.0695	0	0	0
0.5347	0	-1.0695	-1
1.0695	0.8196	-0.5	1

Standardization Formula : $Z = \frac{X-\mu}{\sigma}$

Where:

- X = original value
- μ = mean of the variable
- σ = standard deviation
- Z = standardized value

Large Size Apples (F1)	Rotten Apples (F2)	Damaged Apples (F3)	Small Apples (F4)
1	5	3	1
4	2	6	3
1	4	3	2
4	4	1	1
5	5	2	3

	F1	F2	F3	F4
Mean	3	4	3	2
Std	1.87	1.223	1.87	1

STEP 2: COVARIANCE MATRIX COMPUTATION

Purpose

In this step, we determine **how the variables vary together** with respect to their mean values.

It helps us:

- Identify relationships between variables
- Detect highly correlated features
- Understand the structure of the dataset

It's a symmetric matrix (N×N) that contains:

- Variances on the diagonal
- Covariances off the diagonal

Important Properties

1. $Cov(X, X) = Var(X)$
2. $Cov(X, Y) = Cov(Y, X) \rightarrow$ symmetric
3. Positive value $\rightarrow$ direct correlation
4. Negative value $\rightarrow$ inverse correlation
5. Zero $\rightarrow$ no linear relationship

Covariance Formula

For two variables X and Y :

$$Cov(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Where:

- X_i, Y_i = data values
- $\bar{X}, \bar{Y}$ = means
- n = number of observations

Since the data is already standardized: $\bar{X} = 0$

So, the formula simplifies to:

$$Cov(X, Y) = \frac{1}{n-1} \sum X_i Y_i$$



Covariance Matrix for Two-Dimensional Data

For variables X and Y :

$$\text{Covariance Matrix} = \begin{bmatrix} Var(X) & Cov(X, Y) \\ Cov(Y, X) & Var(Y) \end{bmatrix}$$

	F1	F2	F3	F4
F1	VAR(F1)	COV(F1, F2)	COV(F1, F3)	COV(F1, F4)
F2	COV(F2, F1)	VAR(F2)	COV(F2, F3)	COV(F2, F4)
F3	COV(F3, F1)	COV(F3, F2)	VAR(F3)	COV(F3, F4)
F4	COV(F4, F1)	COV(F4, F2)	COV(F4, F3)	VAR(F4)

Standardized values

F1	F2	F3	F4
-1.0695	0.8196	0	-1
0.5347	-1.6393	1.6042	1
-1.0695	0	0	0
0.5347	0	-1.0695	-1
1.0695	0.8196	-0.5	1

Variance of F1 : $Var(F1) = \frac{1}{n-1} \sum (F1_i - 0)^2$

$$\begin{aligned}
 Var(F1) &= \frac{(-1.0695)^2 + (0.5347)^2 + (-1.0695)^2 + (0.5347)^2 + (1.0695)^2}{5} \\
 &= \frac{1.144 + 0.286 + 1.144 + 0.286 + 1.144}{5} \\
 &= \frac{4.004}{5} = \mathbf{0.78}
 \end{aligned}$$

- Variance of F1 = 0.78 → moderate spread
- Covariance(F1,F2) = -0.8586 → strong negative relationship
→ When F1 increases, F2 tends to decrease

Covariance Between F1 and F2: $Cov(F1,F2) = \frac{1}{n-1} \sum (F1_i)(F2_i)$

Substitute standardized values:

$$\begin{aligned}
 &= \frac{(-1.0695)(0.8196) + (0.5347)(-1.6393) + (-1.0695)(0) + (0.5347)(0) + (1.0695)(0.8196)}{5} \\
 &= \mathbf{-0.8586}
 \end{aligned}$$

Similarly solving all the features, the covariance matrix will be,

- The matrix is symmetric.
- Diagonal values (0.78) are variances.
- Off-diagonal values are covariances.
- Negative values indicate inverse relationships.
- Positive values indicate direct relationships.



	F1	F2	F3	F4
F1	0.78	-0.8586	-0.055	0.424
F2	-0.8586	0.78	-0.607	-0.326
F3	-0.055	-0.607	0.78	0.426
F4	0.424	-0.326	0.426	0.78

STEP 3 & 4: FEATURE VECTOR (Eigenvalues & Eigenvectors)

Purpose

In this step, we find the **principal components**.

Mathematically: $Av = \lambda v$

Where:

- A = covariance matrix
- v = eigenvector
- λ = eigenvalue

To find the Eigenvalues, we solve:

$$\det(A - \lambda I) = 0$$

This is called the **characteristic equation**.

- It expands into a polynomial equation in λ .
- Solving it gives eigenvalues.



Why Eigenvalues Matter?

Eigenvalues = strength of each principal component

- Each eigenvalue represents the **variance captured** by its eigenvector.
- The eigenvector corresponding to the largest eigenvalue becomes:

First *Principal Component* (PC1)



Why Eigenvectors represent?

Each eigenvector:

- Defines a **new direction (axis)** in the data space.
- Is a weighted combination of F1, F2, F3, F4.
- Shows how original features combine to form a principal component.

Example structure:

$$v_1 = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix}$$

Meaning:

$$PC1 = w_1F1 + w_2F2 + w_3F3 + w_4F4$$

Why This Step Is Called "Feature Vector"?

After computing eigenvectors:

We build a matrix:

$$\text{FeatureVector} = [v_1 \ v_2 \ v_3 \ v_4]$$

Each column = one eigenvector.

This matrix will be used to rotate the dataset.



Concept	Meaning
Covariance Matrix	Relationship between features
Eigenvalue	Amount of variance explained
Eigenvector	Direction of maximum variance
Principal Component	Eigenvector ordered by eigenvalue

Subtracted λ from the diagonal

	F1	F2	F3	F4
F1	$0.78 - \lambda$	-0.8586	-0.055	0.424
F2	-0.8586	$0.78 - \lambda$	-0.607	-0.326
F3	-0.055	-0.607	$0.78 - \lambda$	0.426
F4	0.424	-0.326	0.426	$0.78 - \lambda$

$$A - \lambda I =$$

- **characteristic equation** $\det(A - \lambda I) = 0$

Expanding a 4×4 determinant produces a **4th-degree polynomial**:

$$\lambda^4 + a\lambda^3 + b\lambda^2 + c\lambda + d = 0 \iff p(\lambda) = 0$$

When you expand and simplify (very long algebra), you obtain:

$$(\lambda - 2.11691)(\lambda - 0.855413)(\lambda - 0.481689)(\lambda - 0.334007) = 0$$

Solving gives:

$$\lambda = 2.11691, 0.855413, 0.481689, 0.334007$$

- **Solve** $(A - \lambda I)v = 0$

For each eigenvalue:

1. Substitute λ into the matrix
2. Solve the linear system
3. Find a non-zero vector v

Move to
eigenvectors

$$M = \begin{bmatrix} -1.33691 & -0.8586 & -0.055 & 0.424 \\ -0.8586 & -1.33691 & -0.607 & -0.326 \\ -0.055 & -0.607 & -1.33691 & 0.426 \\ 0.424 & -0.326 & 0.426 & -1.33691 \end{bmatrix} \iff Mv = 0$$

Let: $v = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ This gives

$$\begin{aligned} -1.33691x_1 - 0.8586x_2 - 0.055x_3 + 0.424x_4 &= 0 \\ -0.8586x_1 - 1.33691x_2 - 0.607x_3 - 0.326x_4 &= 0 \\ -0.055x_1 - 0.607x_2 - 1.33691x_3 + 0.426x_4 &= 0 \\ 0.424x_1 - 0.326x_2 + 0.426x_3 - 1.33691x_4 &= 0 \end{aligned}$$

Because λ is eigenvalue, determinant is zero $\rightarrow$ system has infinite solutions.....

In reality, we compute eigenvalues as
Python:

```
import numpy as np
np.linalg.eig(A)
```

For $\lambda = 2.11691$ $\longrightarrow$ This means subtracting **2.11691** from each diagonal element of A ($0.78 - 2.11691 = -1.33691$), off-diagonal becomes the same.

We solve:

$$(A - 2.11691I)v = 0$$

After solving the system (using elimination rule), we obtain:

$$v_1 = \begin{bmatrix} 0.515514 \\ -0.616625 \\ 0.399314 \\ 0.441098 \end{bmatrix}$$

This is the **first principal component direction (PC1)**. It corresponds to the largest eigenvalue $\rightarrow$ explains most variance.

Repeat for Remaining Eigenvalues

Doing the same for:

$$\lambda_2 = 0.855413, \lambda_3 = 0.481689, \lambda_4 = 0.334007$$

gives eigenvectors:

$$v_2 = \begin{bmatrix} -0.627 \\ -0.537 \\ -0.455 \\ 0.340 \end{bmatrix}, v_3 = \begin{bmatrix} -0.229 \\ 0.558 \\ -0.785 \\ 0.134 \end{bmatrix}, v_4 = \begin{bmatrix} 0.538 \\ -0.144 \\ -0.122 \\ -0.820 \end{bmatrix}$$

- Eigenvalues tell you **how much variance**
- Eigenvectors tell you **in what direction**

STEP 5: RECAST THE DATA ALONG THE PRINCIPAL COMPONENTS AXES

Purpose

express the data in the **new coordinate system** defined by the principal components.

- 
- Old axes → F1, F2, F3, F4
 - New axes → PC1, PC2, PC3, PC4

Each column = one principal component.

	E1	E2	E3	E4
F1	0.515514	-0.623012	0.0349815	-0.587262
F2	-0.616625	0.113105	0.452326	-0.634336
F3	0.399314	0.744256	-0.280906	-0.455767
F4	0.441098	0.212477	0.845736	0.212173

PC1PC2PC3PC4

Final Data Set=Standardized Data×Feature Vector
Mathematically:

$$Z = X_{std} \cdot V$$

Where:

- 
- X_{std} =standardized dataset (5×4 matrix)
 - V = eigenvector matrix (4×4 matrix)
 - Z = transformed dataset (5×4 matrix)

Standardized Original Data Set

F1	F2	F3	F4
-1.0695	0.8196	0	-1
0.5347	-1.6393	1.6042	1
-1.0695	0	0	0
0.5347	0	-1.0695	-1
1.0695	0.8196	-0.5	1

FeatureVector

E1	E2
0.515514	-0.623012
-0.616625	0.113105
0.399314	0.744256
0.441098	0.212477

$$Z = X_{std} \cdot V$$

$$Z_{reduced} = X_{std} \cdot [E1 \ E2]$$

- Original 4D → becomes 2D
- Maximum variance preserved
- Minimal information loss

The projected dataset after PCA reduction to 2 principal components.

Large Size Apples	Rotten Apples
0.4268066978	0.00116114000000012
-0.4234920968	-0.39182856
-0.551342223	-0.7959219
0.4652040128	0.89996329
0.08272794800000002	0.28653037



The new 5×2 matrix

The percentage of variance

The eigenvalues:

$$\lambda = 2.11691, 0.855413, 0.481689, 0.334007$$

Total variance = sum of all eigenvalues

$$\begin{aligned}\text{Total} &= 2.11691 + 0.855413 + 0.481689 + 0.334007 \\ &= 3.788019\end{aligned}$$

Percentage Explained by Each Component

Formula:

$$\text{Explained Variance Ratio} = \frac{\lambda_i}{\text{Total Variance}} \times 100$$


◆ **PC1**

$$\frac{2.11691}{3.788019} \times 100 = 55.9\%$$

◆ **PC2**

$$\frac{0.855413}{3.788019} \times 100 = 22.6\%$$

◆ **PC3**

$$\frac{0.481689}{3.788019} \times 100 = 12.7\%$$

◆ **PC4**

$$\frac{0.334007}{3.788019} \times 100 = 8.8\%$$

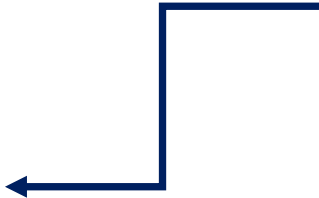
If You Keep Only 2 Components

$$55.9\% + 22.6\% = 78.5\%$$

So, by reducing from **4 features** → **2 principal components**,
you retain:

78.5% of the total variance

That means you kept almost 80% of the information.



Association Rule

What is Market–Basket Analysis?

- A technique used by large retailers to identify associations among items.
- It works by looking for combinations of items that occur frequently in transactions.
- It uses “Association Rules” for analyzing combinations of frequent item sets.

Association Rules

Let's Understand Association Rule...

- It's a technique to know how items are associated to each other.
- **Factors to be considered to measure association:**
 - **Support:** It states how popular an itemset is. It is measured by the proportion of transactions in which the itemset appears.
 - **Confidence:** It states how likely an item Y is purchased when item X is purchased ($X \rightarrow Y$). It is measured by the proportion of transactions containing item X in which item Y also appears.
 - **Lift:** It states how likely an item Y is purchased when item X is purchased, while controlling for how popular item Y is.

Example Transactions

Transaction	Items
1	   
2	  
3	 
4	 
5	   
6	  
7	 
8	 

Total number of transactions = 8

Support Calculation

Support(🍎)

🍎 appears in Transactions: **1, 2, 3, 4** → **4 times**

$$\text{Support}(\{\text{🍎}\}) = \frac{4}{8} = 0.5$$

Support(🍷)

🍷 appears in Transactions:

1, 2, 3, 5, 6, 7 → **6 times** $\text{Support}(\{\text{🍷}\}) = \frac{6}{8} = 0.75$

Support(🍎, 🍷)

Transactions that contain **both** 🍎 and 🍷:
Transactions **1, 2, 3** → **3 times**

$$\text{Support}(\{\text{🍎}, \text{🍷}\}) = \frac{3}{8} = 0.375$$

Confidence Calculation

Confidence (🍎 → 🍷)

$$\begin{aligned} \text{Confidence}(\{\text{🍎} \rightarrow \text{🍷}\}) &= \frac{\text{Support}(\{\text{🍎}, \text{🍷}\})}{\text{Support}(\{\text{🍎}\})} \\ &= \frac{0.375}{0.5} = 0.75 \end{aligned}$$

When customers buy 🍎, 75% of the time they also buy 🍷.

Lift Calculation

Lift (🍎 → 🍷)

$$\begin{aligned} \text{Lift}(\{\text{🍎} \rightarrow \text{🍷}\}) &= \frac{0.375}{0.5 \times 0.75} \\ &= \frac{0.375}{0.375} = 1 \end{aligned}$$

🍎 and 🍷 occur **together exactly as expected by chance**.
There is **no strong association** (Lift = 1).

1	🍎 🍷 🍌
2	🍎 🍷 🍌
3	🍎 🍷 🍌
4	🍎 🍌
5	🍷 🍌 🍌
6	🍷 🍌 🍌
7	🍷 🍌 🍌
8	🍌 🍌



Interpretation of Lift

Lift Value	Meaning
= 1	A and B are independent
> 1	Positive association (A increases chance of B)
< 1	Negative association

$$\text{Lift}(A \Rightarrow B) = P(A \cap B) / P(A) \cdot P(B)$$



$$\text{Lift} = \text{Confidence}(A \Rightarrow B) / \text{Support}(B)$$

LET'S
TRY
IT

Suppose:

- 100 transactions total
- 40 contain Milk
- 30 contain Bread
- 20 contain both

Compute Lift and comment

$$P(\text{Milk}) = \frac{40}{100} = 0.4, P(\text{Bread}) = \frac{30}{100} = 0.3, P(\text{Milk} \cap \text{Bread}) = \frac{20}{100} = 0.2$$

$$\text{Lift} = 0.2 / 0.4 \times 0.3 = 1.67$$

Milk buyers are 1.67 times more likely to also buy Bread than random customers.

- Association rule learning can be divided into three types of algorithms:



Apriori Algorithm

Apriori Algorithm is a basic method used in data analysis to find groups of items that often appear together in large sets of data.

STEP 1 — Set Minimum Support

- Choose a **minimum support threshold** (e.g., 30%).
- Support = frequency of itemset in all transactions.



STEP 2 — Find All Frequent 1-Itemsets (L1)

- Count how many times each item appears.
 - Keep only items whose support $\geq$ minimum support.
- These are your **Frequent 1-itemsets (L1)**.



STEP 3 — Generate Candidate 2-Itemsets (C2)

Combine L1 with itself to generate **all possible 2-item combinations**.

Example: If $L1 = \{A, B, C\}$, then $C2 = \{AB, AC, BC\}$.



STEP 4 — Prune Infrequent Candidates

For every candidate k-itemset:

- Count its support
- Remove (prune) those with support $<$ minimum support

Remaining sets = **Frequent 2-itemsets (L2)**



STEP 5 — Generate Higher-Order Itemsets (k+1)

Use L_k to generate $C(k+1)$:

- Join L_k with L_k to get larger candidate sets
- Example: If $L2$ contains $\{AB, AC\}$, possible $L3$ candidate = ABC
- Prune any candidate whose subsets are not frequent
→ This is the Apriori property at work.

Apriori Algorithm

STEP 6 — Count Support & Prune Again

- Count support for all candidates
 - Keep only those meeting minimum support
 - These become **L3, L4, L5...**
- Stop when no more frequent itemsets can be formed.



STEP 7 — Generate Association Rules

From each frequent itemset:

- Generate rules of form **X → Y**
- Compute:
 - **Confidence**
 - **Lift**
 - (Optionally) **Conviction, Leverage**

Keep rules meeting your thresholds (e.g., Confidence $\geq$ 70%, Lift $\geq$ 1).

- The algorithm makes rules to show how items are related.
- It checks these rules using support, confidence and lift to find the strongest ones.

Association Rule Applications: It has various applications in machine learning and data mining. Below are some popular applications of association rule learning:



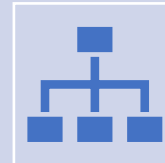
Market Basket Analysis: One of the most common applications of association rule mining, used by large retailers to determine relationships between items purchased together.



Medical Diagnosis: Association rules help identify the probability of certain diseases, making it easier to diagnose and treat patients.



Protein Sequence: Association rules assist in determining the synthesis of artificial proteins.



Catalog Design and Loss-leader Analysis: Association rules are used to optimize catalog layouts and identify products that attract customers, along with many other applications.

THANK
you